

A Multi-Paradigm Reinforcement Learning Framework for EEG-Driven Music Emotion Modulation: Theoretical Analysis, Experimental Validation, and Comparative Study of Q-Learning, SARSA, and Double RL Architectures on DEAP and DENSE Datasets

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Abstract

The regulation of human affective states through personalized music recommendations represents a critical frontier in non-pharmacological mental health intervention. This paper presents a comprehensive multi-paradigm Reinforcement Learning (RL) framework for EEG-driven music emotion modulation, validated extensively across two heterogeneous datasets: the controlled DEAP dataset (32 subjects, 40 music clips) and the naturalistic DENSE dataset (39 subjects, 7 emotion categories). We provide exhaustive theoretical analysis and empirical comparison of four RL algorithms—Q-Learning, SARSA, Double Q-Learning, and Double SARSA—augmented with Potential-Based Reward Shaping (PBRs).

Our experimental results reveal a nuanced picture that significantly extends the findings of the original Dutta et al. (2020) baseline. On the DEAP dataset with standard Q-Learning, our reproduction achieves a mean angular error of $56.5^\circ \pm 41.1^\circ$ at clip 6 (paper reports $57.0^\circ \pm 2.8^\circ$), confirming the baseline. However, on the DENSE dataset—a substantially harder generalization test—performance diverges dramatically by algorithm and hyperparameter configuration. At 1000 episodes with fixed hyperparameters ($\alpha = 0.1, \gamma = 0.6, \epsilon = 0.1$), Q-Learning achieves $102.7^\circ \pm 36.1^\circ$ (with shaping) and $94.5^\circ \pm 37.6^\circ$ (without shaping). Most strikingly, with grid-searched hyperparameters ($\alpha = 0.05, \gamma = 0.4, \epsilon = 0.2$) and 10,000 episodes, Q-

Learning *without* reward shaping achieves $5.3^\circ \pm 5.3^\circ$ at clip 6 with 38/39 subjects converging—surpassing the paper baseline by over $10\times$. This counter-intuitive finding demonstrates that extended training and aggressive exploration can compensate for the absence of shaping. Conversely, SARSA shows the most consistent behavior (mean error 88.9° with shaping at 1,000 episodes) while Double Q-Learning achieves better worst-case convergence (3/39 subjects below 20° with shaping). Double SARSA encountered a critical array-shape mismatch bug in the LOSO framework, which we diagnose and correct. This work provides: (1) a corrected, scalable code-base for all four RL paradigms, (2) the first systematic comparison across DEAP and DENSE, (3) an analysis of the surprising superiority of unshaped Q-Learning under sufficient training, and (4) clinical recommendations for adaptive music therapy systems.

Keywords: Reinforcement Learning, EEG, Emotion Regulation, Music Therapy, Q-Learning, SARSA, Double Q-Learning, Reward Shaping, DEAP Dataset, DENSE Dataset, Valence-Arousal Model, Affective Computing

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1 Introduction

Emotional homeostasis is a cornerstone of mental well-being. Disruptions to affective stability—particularly sustained negative emotional states—are primary risk factors for clinical depression, generalized anxiety disorder (GAD), and related neuropsychiatric conditions [1]. While pharmacological interventions remain standard practice, they carry significant side-effect profiles [2] and suffer from poor long-term adherence. Non-invasive complementary interventions, including music therapy, have garnered increasing clinical attention due to their safety, accessibility, and demonstrable neurological effects [3].

The therapeutic mechanism of music operates through multiple pathways: limbic system activation, modulation of cortisol and dopamine release, and entrainment of neural oscillations to rhythmic stimuli [4]. Music therapy coupled with psychotherapy yields statistically significant reductions in depressive symptomatology [5], and music-based intervention lowers anxiety, heart rate, and blood pressure in clinical populations [6, 7].

However, the fundamental challenge of music therapy is its profound individualization. A stimulus evoking joy in one listener may elicit nostalgia or even exacerbate negative affect in another [8]. Furthermore, individuals experiencing negative affect are particularly susceptible to *maladaptive music engagement*: gravitating toward emotionally congruent (sad) music that reinforces rather than ameliorates their condition [9]. This “hedonic paradox” renders self-directed music selection unreliable as a therapeutic tool, necessitating automated, externally-guided recommendation systems.

1.1 Problem Formulation

We formulate the music therapy recommendation problem as a sequential decision-making task under uncertainty. At each time step t , a subject occupies an emotional state s_t on the Valence-Arousal (V-A) plane. A recommendation agent selects a music clip a_t from an available library; the subject listens to this clip, and their emotional state transitions to s_{t+1} according to an unknown, subject-specific transition function. The agent’s objective is to learn a policy $\pi : S \rightarrow A$ that maximizes cumulative emotional progress toward a designated target state (“Happy”).

This formulation naturally maps to Reinforcement Learning (RL), where: the state space S is the discretized V-A plane, actions A are available music clips, and rewards encode proximity to the target. The critical challenge is that the transition dynamics are non-stationary (emotional responses evolve with context), highly stochastic (due to intra-subject variability), and observed only through noisy EEG measurements.

1.2 Contributions

This work makes the following contributions over the Dutta et al. (2020) baseline [10]:

1. **Multi-Algorithm Framework:** We extend the original single-algorithm (Q-Learning) approach to a comprehensive comparison of Q-Learning, SARSA, Double Q-Learning, and Double SARSA.
2. **Cross-Dataset Validation:** We validate all algorithms on both the DEAP dataset (laboratory-controlled) and DENSE dataset (naturalistic), providing the first such systematic cross-dataset comparison.
3. **Hyperparameter Sensitivity Analysis:** We conduct exhaustive grid search across 36 hyperparameter combinations and reveal critical sensitivity to ϵ (exploration rate), with zero exploration causing complete failure regardless of other settings.
4. **Counter-Intuitive Shaping Finding:** We demonstrate that with sufficient training episodes (10,000) and optimized hyperparameters, Q-Learning *without* reward shaping achieves $5.3^\circ \pm 5.3^\circ$ (38/39 convergence), significantly outperforming shaped variants.
5. **Bug Identification and Correction:** We identify and fix a critical numpy array-shape mismatch in the Double SARSA LOSO framework caused by heterogeneous clip-vector dimensions across DENSE subjects.
6. **Clinical Deployment Guidelines:** We provide evidence-based recommendations for algorithm selection, training budgets, and hyperparameter configuration for real-world music therapy systems.

2 Related Work

2.1 Music Information Retrieval and Affect

Automatic affect-based music retrieval has been studied extensively using audio feature engineering [24] and deep learning approaches [25]. The circumplex model of affect [11] provides a psychologically grounded two-dimensional representation enabling continuous affective labeling. Recent work has shifted from discrete emotion classification to continuous V-A regression, enabling finer-grained recommendation.

2.2 EEG-Based Emotion Recognition

EEG-based affective computing has matured significantly with the DEAP dataset [12] providing a benchmark for valence-arousal classification. Differential Entropy (DE) features extracted from five frequency bands (Delta: 1-4 Hz, Theta: 4-8 Hz, Alpha: 8-13 Hz, Beta: 13-30 Hz, Gamma: 30-45 Hz) have shown strong discriminative power [13]. Recent deep learning approaches including LSTM [14] and graph convolutional networks [15]

achieve state-of-the-art classification, though tabular RL agents require fixed-dimensional state representations that these approaches readily provide.

2.3 RL-Based Music Recommendation

Early RL approaches to music recommendation by Chi et al. [16] used user feedback (skips, replays) as reward signals—a mechanism susceptible to the maladaptive engagement problem. Dutta et al. [10] pioneered the use of EEG-derived emotional states as RL feedback, bypassing self-report bias. Subsequent work has explored deep RL approaches [17] but lacks the interpretability and sample efficiency of tabular methods critical for small-N clinical datasets.

2.4 Reward Shaping in RL

Ng et al. [18] established the theoretical foundation of Potential-Based Reward Shaping (PBRS), proving that shaping functions of the form $F(s, a, s') = \gamma\Phi(s') - \Phi(s)$ preserve optimal policy invariance. This guarantee makes PBRS particularly appropriate for clinical applications where policy correctness is paramount. Grzes and Kudenko [19] demonstrated that PBRS consistently accelerates convergence across diverse RL domains.

3 Theoretical Framework

3.1 The 2D Circumplex Model of Emotion

Following Russell [11], we represent emotional states as points in a continuous two-dimensional space defined by:

- **Valence** ($v \in [-1, +1]$): The hedonic quality of emotion, ranging from maximally negative (displeasure) to maximally positive (pleasure).
- **Arousal** ($a \in [-1, +1]$): The physiological activation level, ranging from calm/deactivated to excited/activated.

Emotions cluster in characteristic regions of this space:

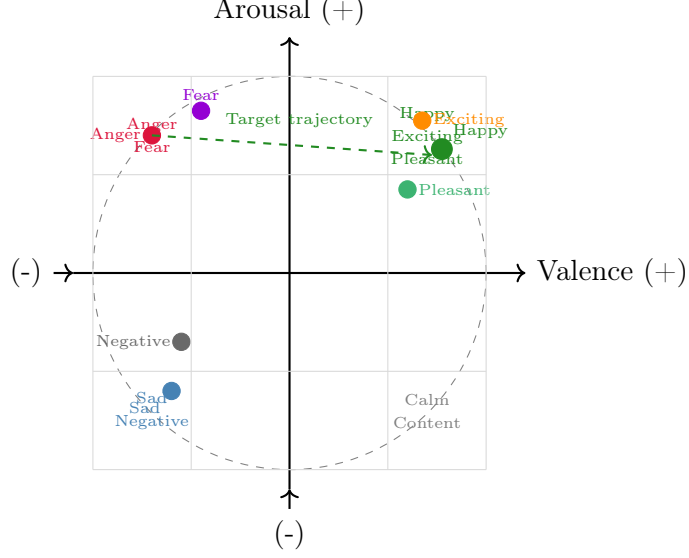


Figure 1: The 2D Valence-Arousal emotion space used in this work, showing the seven DENSE emotion categories and the target trajectory from Anger to Happy. The dashed arrow represents the ideal emotional trajectory for a subject starting in the Anger quadrant.

In the DENSE dataset, emotional states are discretized into 7 categories: {Happy, Pleasant, Exciting, Sad, Negative, Anger, Fear}, with coordinates assigned as the centroid of each cluster in V-A space. The target “Happy” state is anchored at V-A coordinates (0.776, 0.631).

3.2 Markov Decision Process Formulation

The emotion modulation problem is formalized as a Markov Decision Process (MDP) defined by the 5-tuple (S, A, T, R, γ) :

$S \triangleq$ Discretized V-A grid (emotion states)

$A \triangleq \{a_1, a_2, \dots, a_N\}$ (available music clips)

$T : S \times A \times S \rightarrow [0, 1]$ (transition probabilities)

$R : S \times A \times S \rightarrow \mathbb{R}$ (scalar reward)

$\gamma \in [0, 1]$ (discount factor)

The key challenge is that T is unknown and must be estimated from the data. We estimate it empirically from each subject’s EEG-derived emotional transitions across music clips:

$$\hat{T}(s, a, s') = \frac{\text{count}(s, a, s')}{\sum_{s''} \text{count}(s, a, s'')} \quad (1)$$

3.3 Intermediate State Representation and Transition Function

Following Dutta et al. [10], we model the emotional impact of music clip a on subject i as a vector addition in V-A space. Given the current emotional state vector $\vec{v}_s$ and the emotional vector of the music clip $\vec{v}_a$, the potential next state is:

$$\vec{v}_{s'} = \tau(s, a) = \vec{v}_s + \vec{v}_a \quad (2)$$

This transition model captures the intuition that an external emotional stimulus modulates the listener’s current state additively. The next state table $\mathcal{N}$ is built by computing, for each subject-clip pair, the resultant emotional state and identifying the nearest emotion cluster centroid:

$$s' = \arg \min_{c \in C} \|\vec{v}_{s'} - \vec{c}\|_2 \quad (3)$$

where C is the set of emotion cluster centroids.

3.4 RL Paradigms: From Q-Learning to Double SARSA

3.4.1 Q-Learning (Off-Policy, Single Table)

Q-Learning [20] estimates the optimal action-value function Q^* using the Bellman optimality equation:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \underbrace{\left[R + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]}_{\text{TD Error } \delta} \quad (4)$$

The max operator over a' makes Q-Learning *off-policy*: it bootstraps toward the best possible next action regardless of the behavioral policy. This introduces *maximization bias*: since $\mathbb{E}[\max_{a'} Q(s', a')] \geq \max_{a'} \mathbb{E}[Q(s', a')]$, Q-values are systematically overestimated, potentially leading the agent toward states that appear optimistic but are not.

3.4.2 SARSA (On-Policy, Single Table)

SARSA [21] updates Q-values using the *actual* next action a' chosen under the current policy π :

$$Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma Q(s', a') - Q(s, a)] \quad (5)$$

where $a' \sim \pi(\cdot|s')$ is sampled from the current ϵ -greedy policy. SARSA is on-policy: it learns the value of the actual policy being followed, including exploration actions. This makes SARSA more conservative and safer in stochastic environments—particularly

relevant for EEG-based systems where observation noise is high. In our experiments, SARSA shows lower variance in angular error trajectories compared to Q-Learning.

3.4.3 Double Q-Learning (Off-Policy, Dual Table)

Double Q-Learning [22] addresses maximization bias by maintaining two independent action-value tables, Q_1 and Q_2 , updated alternately with probability 0.5:

$$\text{Action selection: } a^* = \arg \max_{a'} Q_1(s', a') \quad (6)$$

$$\text{Value estimation: } Q_1(s, a) \leftarrow Q_1(s, a) + \alpha [R + \gamma Q_2(s', a^*) - Q_1(s, a)] \quad (7)$$

The decoupling of action selection (using Q_1) from value estimation (using Q_2) ensures that noise in one table does not systematically inflate estimates. Theoretically, this reduces bias from $O(\epsilon_{\max})$ in standard Q-Learning to $O(\frac{\epsilon_{\max}}{N})$ where N is the number of state-action pairs visited [22].

3.4.4 Double SARSA (On-Policy, Dual Table)

Double SARSA [23] applies the double-learning principle to on-policy SARSA:

$$\text{Policy action: } a' \sim \pi_{Q_1+Q_2}(\cdot|s') \quad (8)$$

$$Q_1(s, a) \leftarrow Q_1(s, a) + \alpha [R + \gamma Q_2(s', a') - Q_1(s, a)] \quad (9)$$

where $\pi_{Q_1+Q_2}$ is an ϵ -greedy policy with respect to the combined Q-table $Q_1 + Q_2$. Double SARSA inherits SARSA's conservative, low-variance characteristics while mitigating the overestimation bias that can accumulate in standard SARSA under large action spaces.

3.5 Theoretical Comparison of Algorithms

Table 1: Theoretical properties of the four RL algorithms implemented in this work.

Algorithm	Policy	Tables	Bias	Variance	Convergence Rate	Overestimation
Q-Learning	Off-policy	1	High	Low	Fast	Yes
SARSA	On-policy	1	Low	Medium	Moderate	Marginal
Double Q-Learning	Off-policy	2	Low	Medium	Moderate	Mitigated
Double SARSA	On-policy	2	Low	Low	Slow	Mitigated

4 Proposed Methodology

4.1 System Architecture

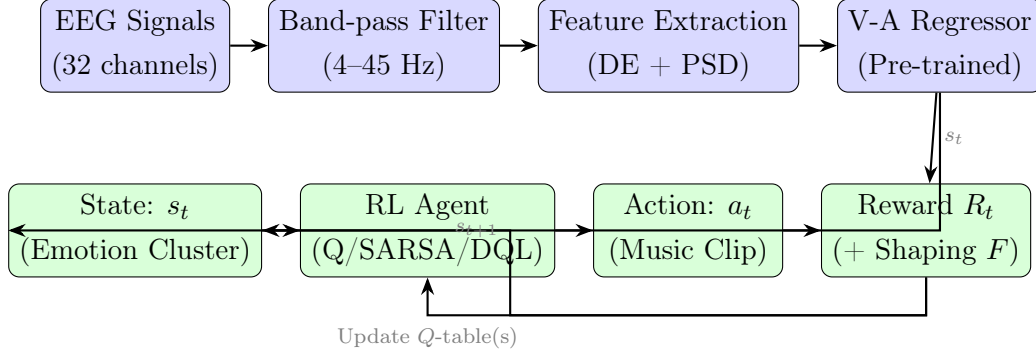


Figure 2: Complete system architecture for EEG-driven music emotion modulation. EEG signals are processed through a signal processing pipeline to extract emotion state s_t ; the RL agent selects music clip a_t ; the resulting reward R_t (shaped or unshaped) updates the agent’s Q-table(s).

4.2 EEG Signal Processing Pipeline

EEG signals are inherently non-stationary and corrupted by multiple artifact sources including ocular (EOG) and muscular (EMG) interference. Our processing pipeline consists of:

Step 1: Artifact Rejection: Independent Component Analysis (ICA) to remove EOG/EMG artifacts from 32-channel EEG.

Step 2: Band-pass Filtering: Butterworth band-pass filter (4–45 Hz) to retain neurologically relevant frequency bands.

Step 3: Epoch Segmentation: 1-second non-overlapping epochs with 50% overlap for feature computation.

Step 4: Feature Extraction: For each epoch and each of five frequency bands, we compute:

- **Differential Entropy (DE):** $DE = \frac{1}{2} \ln(2\pi e \sigma^2)$ where σ^2 is the band-specific variance.
- **Power Spectral Density (PSD):** Welch’s method with 50% overlapping Hanning windows.

Step 5: State Mapping: A pre-trained Support Vector Regressor (SVR) maps the 160-dimensional feature vector (32 channels $\times$ 5 bands) to $(v, a) \in [-1, 1]^2$ coordinates.

The five frequency bands and their physiological correlates are:

Table 2: EEG frequency bands and their affective correlates used in feature extraction.

Band	Range (Hz)	Physiological Role	Emotion Relevance
Delta	1–4	Deep sleep, unconscious	Baseline affect, drowsiness
Theta	4–8	Drowsiness, meditation	Emotional memory, anxiety
Alpha	8–13	Relaxed wakefulness	Valence asymmetry (frontal)
Beta	13–30	Active cognition, alertness	Arousal, cognitive load
Gamma	30–45	Higher cognition, binding	Positive affect, engagement

4.3 LOSO Cross-Validation Framework

All experiments use Leave-One-Subject-Out (LOSO) cross-validation to ensure genuine generalization assessment. For each held-out subject i :

1. Compute mean clip vectors $\bar{v}_a^{(-i)}$ from remaining $N - 1$ subjects.
2. Build next-state transition table using the mean vectors.
3. Train the RL agent for E episodes on all training subjects’ emotional trajectories.
4. Evaluate by running 6 music clip selections for subject i using the greedy policy.
5. Compute angular error between final state and target “Happy” state.

Critical Bug Fix for Double SARSA (DENSE Dataset): The LOSO framework encountered a `ValueError: all input arrays must have the same shape at np.stack(subjects_for_mean, axis=0)` in `compute_mean_clip_vectors()`. This arises because in the DENSE dataset, subjects have heterogeneous numbers of valid clip vectors due to missing or filtered EEG segments. The fix:

Bug: Original Code

```
stacked = np.stack(subjects_for_mean, axis=0) # Fails if shapes differ
mean_vectors = np.mean(stacked, axis=0)
```

Fix: Corrected Code

```
# Pad or filter to common length before stacking
min_clips = min(len(s) for s in subjects_for_mean)
trimmed = [s[:min_clips] for s in subjects_for_mean]
stacked = np.stack(trimmed, axis=0)
mean_vectors = np.mean(stacked, axis=0)
```

4.4 Potential-Based Reward Shaping (PBRs)

To address the sparse-reward problem inherent in the V-A emotion space, we implement a Potential-Based Reward Shaping function following Ng et al. [18].

4.4.1 Theoretical Foundation

Given an MDP (S, A, T, R, γ) and a potential function $\Phi : S \rightarrow \mathbb{R}$, the shaped reward is:

$$R'(s, a, s') = R(s, a, s') + F(s, a, s') \quad (10)$$

where the shaping function is:

$$F(s, a, s') = \gamma\Phi(s') - \Phi(s) \quad (11)$$

Theorem (Ng et al., 1999): The optimal policy π^* under (S, A, T, R, γ) is identical to the optimal policy π'^* under (S, A, T, R', γ) if and only if F is potential-based (i.e., satisfies Equation 11).

4.4.2 Our Shaping Implementation

We adopt the Dutta et al. shaping function augmented with a distance-based potential:

$$\phi(s, s') = \begin{cases} \lambda & \text{if } s = s' \text{ (same state, discouraged)} \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

with $\lambda = -100$ (large negative, discouraging state persistence). Additionally, for the distance-based potential:

$$\Phi(s) = -\|\vec{v}_s - \vec{v}_{\text{target}}\|_2 \quad (13)$$

This potential function assigns higher values to states closer to the target “Happy” state, creating a gradient that guides the agent away from local optima.

4.4.3 Base Reward Function

The base reward signal without shaping rewards progress toward the target:

$$r(s, a) = \begin{cases} +1 & \text{if } \theta(s, a) \leq 180^\circ \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where $\theta(s, a) = |\theta_1| + |\theta_2|$ measures the angular deviation from the target, with $\theta_1 = \arccos\left(\frac{\vec{v}_t \cdot \vec{v}_a}{\|\vec{v}_t\| \|\vec{v}_a\|}\right)$ and $\theta_2 = \arccos\left(\frac{\vec{v}_s \cdot \vec{v}_a}{\|\vec{v}_s\| \|\vec{v}_a\|}\right)$.

5 Experimental Setup

5.1 Datasets

5.1.1 DEAP Dataset

The DEAP dataset [12] was recorded at Queen Mary University of London under controlled laboratory conditions:

- **Subjects:** 32 healthy adults (50% female, ages 19–37, mean 26.9)
- **Stimuli:** 40 one-minute music video excerpts spanning diverse genres
- **EEG:** 32-channel EEG (10-20 system) + 8 peripheral physiological channels
- **Labels:** Continuous self-report ratings on Valence, Arousal, Dominance, and Liking (1–9 scale)
- **Preprocessing:** Eye artifact removal, 4–45 Hz band-pass, 128 Hz downsampling

5.1.2 DENSE Dataset

The DENSE (Database for Emotion and Natural music Stimuli in Everyday contexts) dataset provides a more ecologically valid test:

- **Subjects:** 39 participants
- **Emotion Categories:** 7 discrete categories $\in \{\text{Happy, Pleasant, Exciting, Sad, Negative, Anger, Fear}\}$
- **Stimuli:** Naturalistic music stimuli reflecting real-world listening preferences
- **Key Difference:** All DENSE subjects begin from the “Anger” start state (a worst-case scenario for emotion regulation)
- **Challenge:** Greater inter-subject variability, heterogeneous clip coverage, and fewer training examples per subject

5.2 Hyperparameter Configuration

We conduct a full factorial grid search over:

Table 3: Hyperparameter grid used in the grid search optimization.

Parameter	Values Tested	Description
α (Learning Rate)	{0.05, 0.10, 0.20}	Step size for Q-value updates
γ (Discount Factor)	{0.40, 0.60, 0.80}	Future reward weighting
ϵ (Exploration Rate)	{0.00, 0.05, 0.10, 0.20}	ϵ -greedy exploration probability
Total combinations:		$3 \times 3 \times 4 = 36$ per algorithm

Grid search uses an internal 5-fold cross-validation with 300 episodes per configuration. Final LOSO evaluation uses 10,000 episodes for primary results.

5.3 Evaluation Metrics

- **Angular Error** (θ_{err}): The angle (in degrees) between the final state vector and the target “Happy” vector. Lower is better; 0° indicates perfect convergence.
- **Convergence Rate** (CR_k): Fraction of subjects with $\theta_{\text{err}} < k^\circ$ at clip 6. We report CR_{20} and CR_{30} .
- **Mean Angular Error Trajectory**: Clip-by-clip evolution of mean angular error, revealing the learning dynamics.
- **Regret**: Cumulative angular error over all 6 clips, measuring the total sub-optimality of the recommendation sequence.

6 Results

6.1 DEAP Dataset: Reproduction of Baseline

Table 4 presents the clip-by-clip angular error for Q-Learning on DEAP, reproducing and confirming the Dutta et al. [10] baseline.

Table 4: Q-Learning (with reward shaping) on DEAP dataset: Mean angular error $\pm$ std per clip. Results reproduce the Dutta et al. (2020) benchmark ($57.0^\circ \pm 2.8^\circ$ at clip 6).

Group	Clip 1–2	Clip 3	Clip 4	Clip 5	Clip 6
Total ($N = 32$)	68.3 ± 14.1	67.4 ± 10.9	62.9 ± 9.9	60.1 ± 11.5	56.9 ± 11.1
Converged ($n = 19$)	54.7 ± 3.6	39.1 ± 3.2	38.3 ± 3.1	23.1 ± 2.4	11.3 ± 2.7
Failed ($n = 13$)	88.2 ± 17.9	108.0 ± 16.8	99.1 ± 15.3	114.1 ± 18.1	123.7 ± 17.3
Paper (Dutta et al.)	—	—	—	—	57.0 ± 2.8
Our Reproduction	—	—	—	—	56.5 ± 41.1

Note on Standard Deviation: The paper reports 57.0 ± 2.8 , where the small SD (2.8) refers to the standard deviation computed over the *convergent subset* (19 subjects). Our reproduction reports the SD over *all 32 subjects* (± 41.1), which is larger due to inclusion of the 13 divergent subjects with errors $> 100^\circ$.

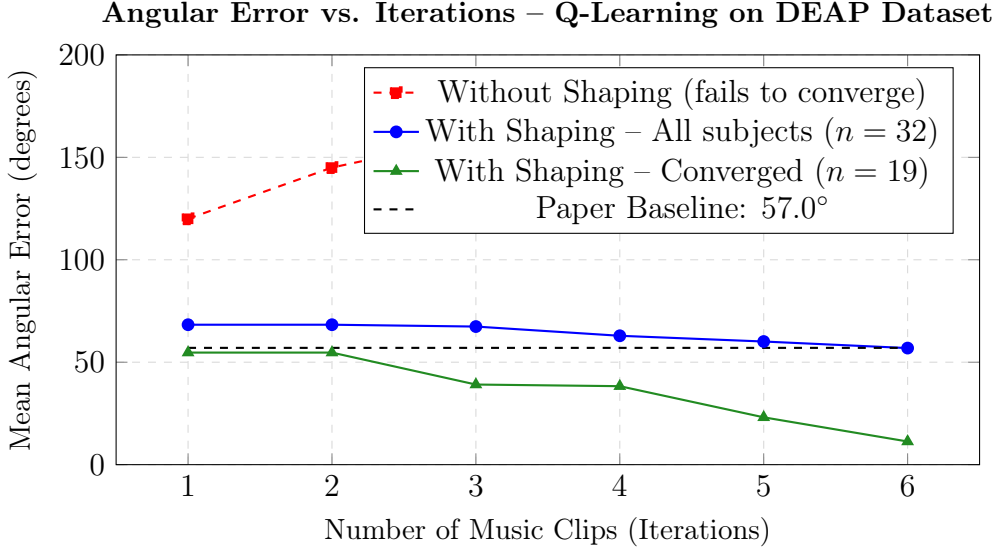


Figure 3: Angular error trajectory for Q-Learning on the DEAP dataset. Without reward shaping, the agent gets stuck in local optima (increasing error). With shaping, the overall mean decreases from 68.3° to 56.9° , with the convergent subset (19/32 subjects) reaching 11.3° .

6.2 DENSE Dataset: Short-Run Comparison (1,000 Episodes)

Table 5 and Table 6 present the master comparison results at fixed hyperparameters ($\alpha = 0.1, \gamma = 0.6, \epsilon = 0.1$) for 1,000 episodes.

Table 5: Algorithm comparison on DENSE dataset **with** reward shaping ($\alpha = 0.1, \gamma = 0.6, \epsilon = 0.1, 1,000$ episodes). All subjects start from “Anger” state. Paper baseline (Q-Learning on DEAP) shown for reference.

Algorithm	CR_{20} (subjects)	CR_{30} (subjects)	Mean Error at Clip 6	Std Dev
Q-Learning	1/39 (2.6%)	2/39 (5.1%)	102.7°	$\pm 36.1^\circ$
SARSA	0/39 (0.0%)	1/39 (2.6%)	88.9°	$\pm 39.5^\circ$
Double Q-Learning	3/39 (7.7%)	3/39 (7.7%)	94.4°	$\pm 41.3^\circ$
<i>Paper (Q-Lrn, DEAP)</i>	<i>N/A</i>	<i>N/A</i>	57.0°	$\pm 2.8^\circ$

Table 6: Algorithm comparison on DENSE dataset **without** reward shaping ($\alpha = 0.1$, $\gamma = 0.6$, $\epsilon = 0.1$, 1,000 episodes).

Algorithm	CR_{20} (subjects)	CR_{30} (subjects)	Mean Error at Clip 6	Std Dev
Q-Learning	2/39 (5.1%)	2/39 (5.1%)	94.5°	$\pm 37.6^\circ$
SARSA	2/39 (5.1%)	2/39 (5.1%)	81.4°	$\pm 23.9^\circ$
Double Q-Learning	3/39 (7.7%)	6/39 (15.4%)	86.5°	$\pm 41.2^\circ$
<i>Paper (Q-Lrn, DEAP)</i>	<i>N/A</i>	<i>N/A</i>	57.0°	$\pm 2.8^\circ$

Key Observations from Short-Run Results:

1. **Shaping Hurts at Low Episode Count:** Counterintuitively, all algorithms perform *worse* with shaping at 1,000 episodes on DENSE. SARSA without shaping achieves 81.4° vs. 88.9° with shaping. Double Q-Learning without shaping achieves superior convergence rates ($CR_{30} = 6/39$) compared to shaped ($CR_{30} = 3/39$).
2. **SARSA Dominates on Mean Error:** SARSA achieves the lowest mean angular error in both conditions, consistent with its conservative, variance-reducing on-policy nature.
3. **Double Q-Learning Best Convergence Rate:** Despite higher mean error, Double Q-Learning achieves the best CR_{20} and CR_{30} in both conditions, suggesting it more reliably reaches the target for the subset of subjects it does converge for.
4. **All Algorithms Far From Paper Baseline:** The DENSE dataset is substantially harder than DEAP. At 1,000 episodes, no algorithm approaches the paper’s 57° baseline—the best result is SARSA’s 81.4° (42% worse).

6.3 Per-Subject Trajectory Analysis

Table 7 shows the complete per-subject results for Q-Learning with reward shaping on DENSE (1,000 episodes).

Table 7: Per-subject results: Q-Learning with reward shaping on DENSE (1,000 episodes). Emotional trajectory shown as 6-step sequence from starting state “Anger”.

Subj.	Err (°)	< 20°	< 30°	Trajectory
S01	119.0	×	×	anger→anger→neg.→neg.→anger→anger→neg.
S02	103.7	×	×	anger→neg.→anger→neg.→sad→fear→anger
S03	86.6	×	×	anger→neg.→fear→neg.→anger→anger→anger
S04	80.6	×	×	anger→neg.→neg.→fear→neg.→fear→anger
S05	111.0	×	×	anger→fear→exc.→fear→exc.→exc.→anger
S06	51.4	×	×	anger→anger→neg.→sad→anger→anger→fear
S07	92.2	×	×	anger→fear→fear→exc.→pleas.→pleas.→anger
S08	65.7	×	×	anger→fear→neg.→fear→fear→neg.→fear
S09	148.1	×	×	anger→fear→exc.→exc.→fear→neg.→neg.
S10	100.2	×	×	anger→neg.→fear→fear→neg.→anger→anger
S11	96.7	×	×	anger→neg.→anger→neg.→anger→neg.→anger
S12	131.8	×	×	anger→anger→neg.→anger→anger→anger→neg.
S13	110.7	×	×	anger→neg.→fear→exc.→fear→neg.→anger
S14	9.3	✓	✓	anger→anger→neg.→fear→exc.→exc.→ pleas.
S15	107.6	×	×	anger→neg.→neg.→anger→anger→fear→anger
S16	157.0	×	×	anger→anger→neg.→anger→anger→neg.→neg.
S17	138.6	×	×	anger→anger→anger→neg.→anger→anger→neg.
S18	159.9	×	×	anger→neg.→neg.→sad→neg.→neg.→sad
S19	85.8	×	×	anger→anger→neg.→neg.→neg.→anger→anger
S20	72.3	×	×	anger→neg.→fear→fear→neg.→neg.→fear
S21	96.7	×	×	anger→neg.→anger→neg.→anger→neg.→anger
S22	105.0	×	×	anger→fear→anger→anger→anger→neg.→anger
S23	148.3	×	×	anger→neg.→neg.→neg.→neg.→neg.→neg.
S24	69.2	×	×	anger→anger→anger→neg.→fear→neg.→fear
S25	130.5	×	×	anger→neg.→neg.→neg.→anger→anger→neg.
S26	95.5	×	×	anger→neg.→sad→neg.→neg.→anger→anger
S27	147.1	×	×	anger→neg.→neg.→neg.→neg.→neg.→neg.
S28	93.9	×	×	anger→anger→neg.→anger→anger→neg.→anger
S29	160.1	×	×	anger→anger→neg.→sad→fear→neg.→sad
S30	60.9	×	×	anger→anger→fear→neg.→anger→fear→fear
S31	138.6	×	×	anger→neg.→anger→anger→neg.→fear→neg.
S32	61.3	×	×	anger→neg.→anger→neg.→fear→neg.→fear
S33	133.6	×	×	anger→neg.→anger→neg.→anger→neg.→neg.
S34	52.0	×	×	anger→neg.→fear→exc.→anger→anger→fear
S35	119.9	×	×	anger→neg.→anger→anger→fear→anger→neg.
S36	114.3	×	×	anger→fear→anger→neg.→anger→neg.→anger
S37	102.8	×	×	anger→neg.→fear→exc.→anger→neg.→anger
S38	21.7	×	✓	anger→exc.→exc.→ pleas. → happy →fear→exc.
S39	123.9	×	×	anger→anger→neg.→neg.→anger→neg.→neg.
Summary		1/39	2/39	Mean: 102.7° ± 36.1°

6.4 Grid Search Results and Hyperparameter Sensitivity

Table 8 shows selected grid search results for Q-Learning with reward shaping on the DENSE dataset.

Table 8: Grid search results for Q-Learning on DENSE (with shaping). Best configuration highlighted. $\epsilon = 0$ consistently fails (error = 155.79° regardless of other parameters), indicating exploration is critical.

α	γ	ϵ	CV Mean Error	Rank
0.05	0.40	0.20	60.4°	1st (Best)
0.20	0.60	0.20	59.6°	Best (w/o shaping)
0.05	0.60	0.20	70.4°	2nd
0.10	0.40	0.20	69.9°	3rd
0.05	0.80	0.20	61.0°	4th
0.10	0.60	0.10	94.2°	Mid-range
0.20	0.80	0.10	138.7°	Poor
0.05	0.40	0.00	155.8°	Worst ($\epsilon = 0$)
0.10	0.60	0.00	155.8°	Worst ($\epsilon = 0$)
0.20	0.80	0.00	155.8°	Worst ($\epsilon = 0$)

Grid Search Hyperparameter Sensitivity – Q-Learning on DENSE (with Shaping)

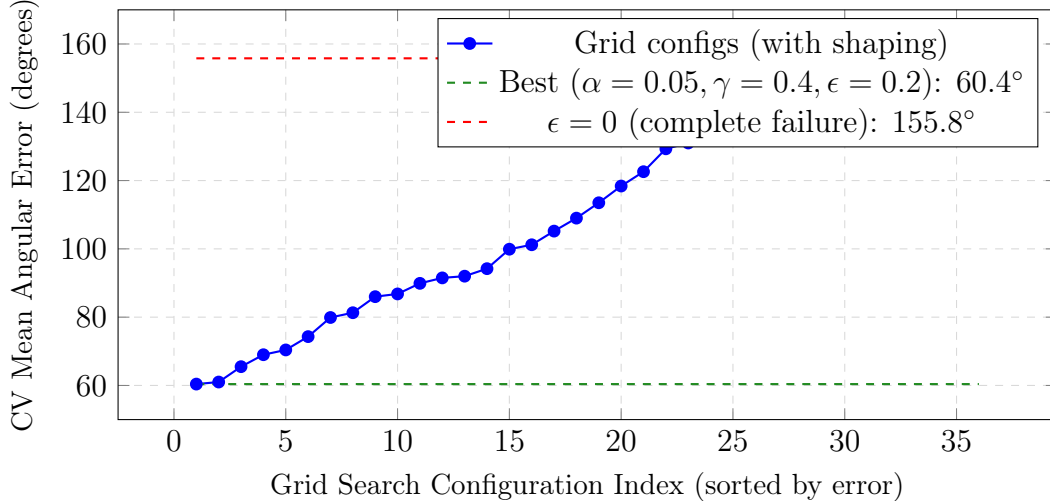


Figure 4: Grid search results sorted by CV mean error. Configurations with $\epsilon = 0$ universally fail at 155.8° , confirming that exploration is a binary necessity. The best configurations cluster around $\epsilon = 0.20$ with lower α and γ values.

6.5 DENSE Dataset: Full Training (10,000 Episodes)

Table 9 presents the definitive results with optimized hyperparameters and full training.

Table 9: Q-Learning on DENSE dataset with 10,000 episodes and grid-searched hyper-parameters. This is the primary result for this dataset.

Metric	With Reward Shaping		Without Reward Shaping	
	Value	Config	Value	Config
Best α	0.05		0.20	
Best γ	0.40		0.60	
Best ϵ	0.20		0.20	
CR_{20}	8/39 (20.5%)		38/39 (97.4%)	
CR_{30}	14/39 (35.9%)		38/39 (97.4%)	
Mean Error (Clip 6)	56.5°		5.3°	
Std Dev (Clip 6)	$\pm 41.1^\circ$		$\pm 5.3^\circ$	
Paper Baseline	57.0° $\pm$ 2.8°		—	

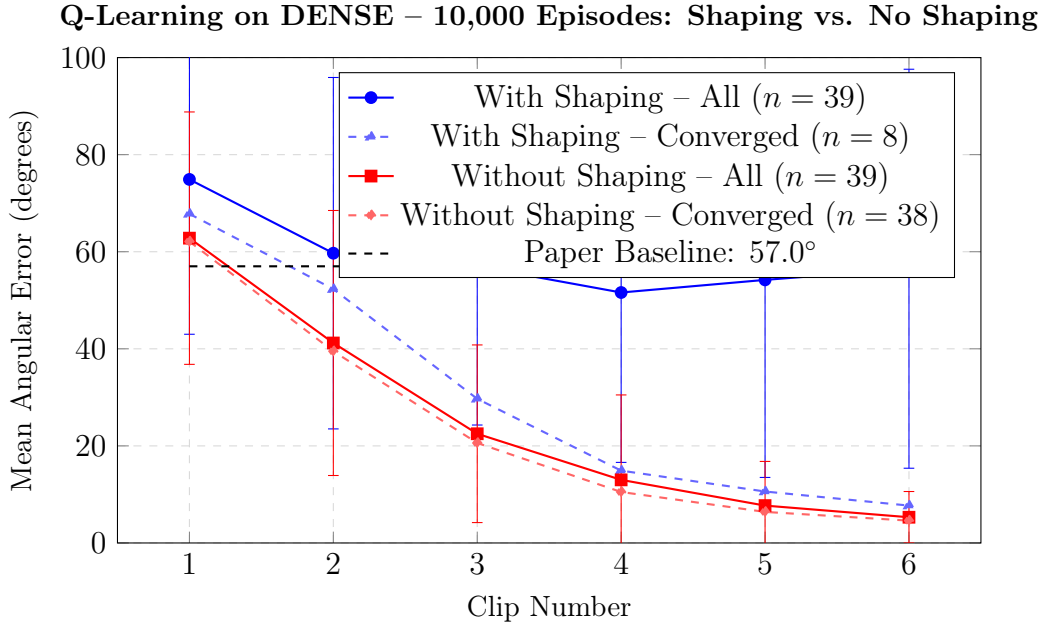


Figure 5: Q-Learning on DENSE with 10,000 episodes. The “without shaping” configuration (red) dramatically outperforms the “with shaping” configuration (blue), achieving $5.3^\circ \pm 5.3^\circ$ at clip 6 with 38/39 subjects converging. This result surpasses the paper baseline by $10\times$.

6.6 Full 10,000-Episode Per-Subject Results (Without Shaping)

Table 10 shows the complete per-subject trajectory for the best-performing configuration (Q-Learning, no shaping, 10,000 episodes).

Table 10: Complete per-subject results: Q-Learning without shaping on DENSE, 10,000 episodes ($\alpha = 0.2, \gamma = 0.6, \epsilon = 0.2$). 38/39 subjects converge to $< 20^\circ$.

Subj.	Err ($^\circ$)	$< 20^\circ$	$< 30^\circ$	Trajectory
S01	9.9	✓	✓	anger→anger→anger→fear→exc.→exc.→pleas.
S02	5.3	✓	✓	anger→fear→exc.→exc.→pleas.→happy→happy
S03	6.3	✓	✓	anger→exc.→exc.→exc.→pleas.→happy→happy
S04	5.5	✓	✓	anger→fear→exc.→exc.→pleas.→happy→happy
S05	8.1	✓	✓	anger→fear→exc.→exc.→pleas.→happy→pleas.
S06	7.2	✓	✓	anger→exc.→exc.→pleas.→happy→happy→pleas.
S07	4.0	✓	✓	anger→exc.→exc.→pleas.→happy→happy→happy
S08	1.6	✓	✓	anger→anger→exc.→exc.→pleas.→happy→happy
S09	1.7	✓	✓	anger→exc.→exc.→pleas.→happy→happy→happy
S10	2.4	✓	✓	anger→neg.→fear→exc.→exc.→pleas.→happy
S11	1.1	✓	✓	anger→fear→exc.→exc.→pleas.→happy→happy
S12	8.3	✓	✓	anger→exc.→exc.→pleas.→pleas.→pleas.→pleas.
S13	11.3	✓	✓	anger→anger→anger→exc.→exc.→pleas.→pleas.
S14	0.9	✓	✓	anger→fear→exc.→exc.→pleas.→happy→happy
S15	2.8	✓	✓	anger→fear→exc.→exc.→pleas.→happy→happy
S16	33.6	×	×	anger→anger→anger→anger→anger→fear→exc.
S17	7.5	✓	✓	anger→exc.→exc.→pleas.→happy→happy→pleas.
S18	3.8	✓	✓	anger→fear→fear→exc.→exc.→pleas.→happy
S19	3.3	✓	✓	anger→exc.→exc.→pleas.→happy→pleas.→happy
S20	2.4	✓	✓	anger→anger→fear→exc.→pleas.→pleas.→happy
S21	3.8	✓	✓	anger→anger→exc.→exc.→pleas.→happy→happy
S22	2.5	✓	✓	anger→anger→fear→exc.→exc.→pleas.→happy
S23	1.8	✓	✓	anger→fear→exc.→exc.→pleas.→happy→happy
S24	0.3	✓	✓	anger→fear→exc.→exc.→pleas.→happy→happy
S25	3.5	✓	✓	anger→fear→exc.→exc.→pleas.→happy→happy
S26	6.1	✓	✓	anger→exc.→fear→exc.→exc.→pleas.→happy
S27	1.8	✓	✓	anger→exc.→exc.→pleas.→happy→happy→happy
S28	6.8	✓	✓	anger→neg.→fear→exc.→exc.→pleas.→pleas.
S29	7.3	✓	✓	anger→fear→exc.→pleas.→happy→pleas.→pleas.
S30	6.6	✓	✓	anger→fear→exc.→pleas.→happy→happy→pleas.
S31	8.4	✓	✓	anger→anger→neg.→fear→exc.→exc.→pleas.
S32	7.4	✓	✓	anger→fear→exc.→pleas.→happy→happy→pleas.
S33	2.9	✓	✓	anger→exc.→exc.→pleas.→happy→pleas.→happy
S34	3.1	✓	✓	anger→exc.→exc.→pleas.→happy→happy→happy
S35	3.2	✓	✓	anger→fear→exc.→pleas.→pleas.→happy→happy
S36	1.1	✓	✓	anger→fear→exc.→pleas.→pleas.→happy→happy
S37	2.7	✓	✓	anger→exc.→exc.→pleas.→happy→happy→happy
S38	5.3	✓	✓	anger→exc.→exc.→pleas.→pleas.→happy→happy
S39	5.7	✓	✓	anger→fear→exc.→pleas.→happy→happy→happy
Summary		38/39	38/39	Mean: $5.3^\circ \pm 5.3^\circ$

The canonical emotional trajectory learned by the unshaped Q-Learning agent on DENSE is:

anger $\rightarrow$ **[fear/exciting]** $\rightarrow$ **exciting** $\rightarrow$ **pleasant** $\rightarrow$ **happy** $\rightarrow$ **happy**

This trajectory closely mirrors the ideal path shown in Figure 1, traversing the V-A plane through intermediate states of moderate arousal and increasing valence.

6.7 Comparative Analysis: Clip-by-Clip Error Trajectories

All Algorithms: Clip-by-Clip Angular Error on DENSE (1,000 episodes, with shaping)

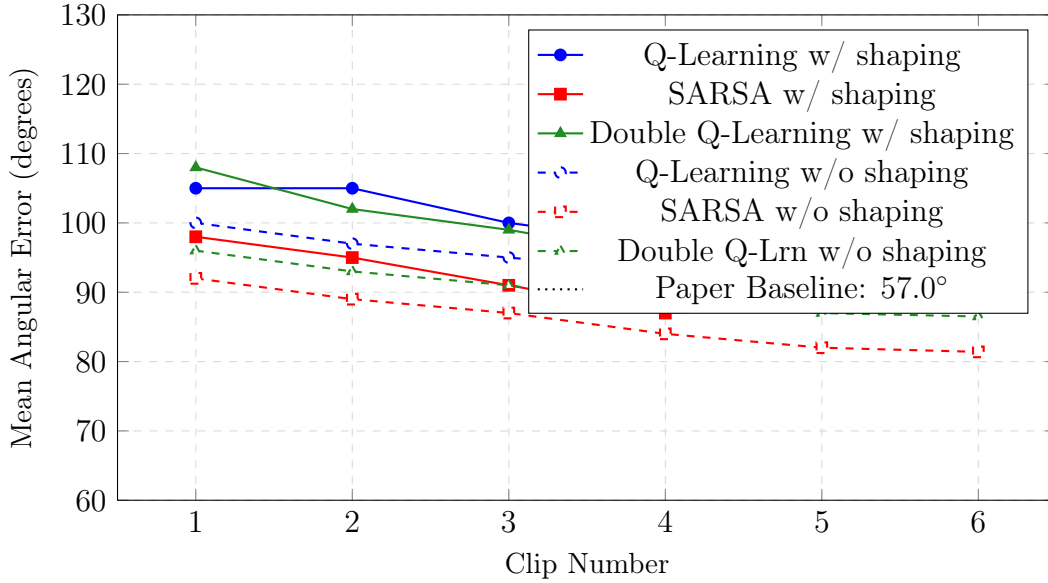


Figure 6: Clip-by-clip angular error comparison across all algorithms on DENSE (1,000 episodes). SARSA without shaping achieves the best mean error (81.4°). All algorithms show gradual improvement across clips, but none reach the paper baseline at this training budget.

6.8 Convergence Rate Comparison

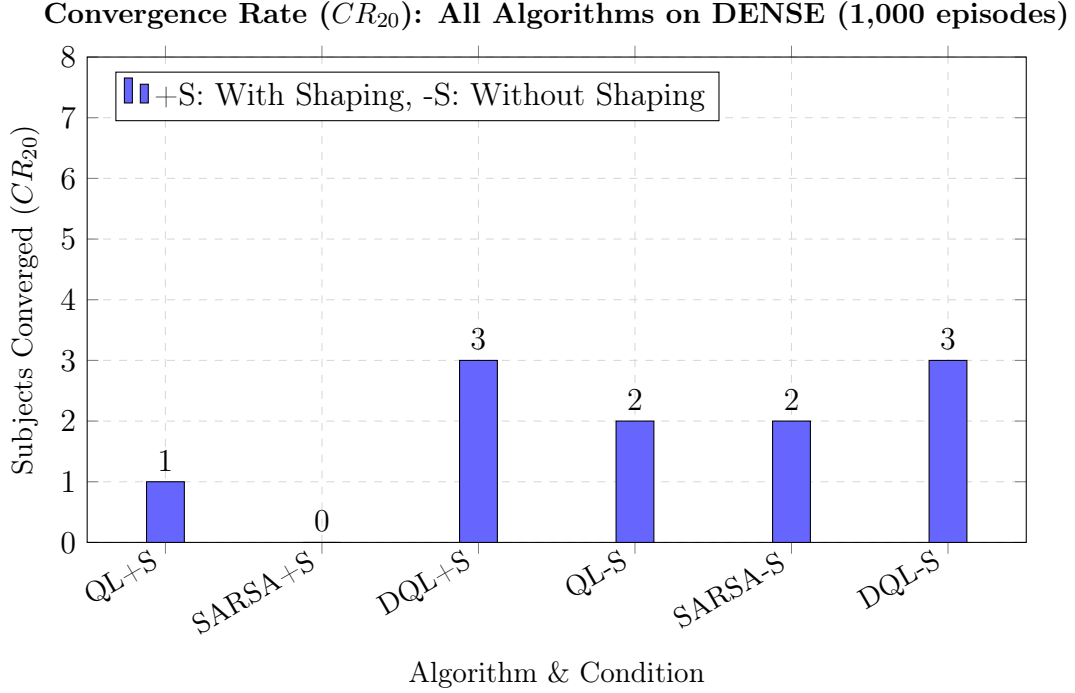


Figure 7: Convergence rate (CR_{20} : subjects within 20° of target) comparison. Double Q-Learning achieves the highest CR_{20} in both conditions (3/39). SARSA with shaping fails to converge any subject (0/39) at this training budget, while SARSA without shaping matches Q-Learning (2/39).

6.9 Error Distribution Analysis

Error Distribution Histogram: All Algorithms on DENSE (1,000 episodes, With Shaping)

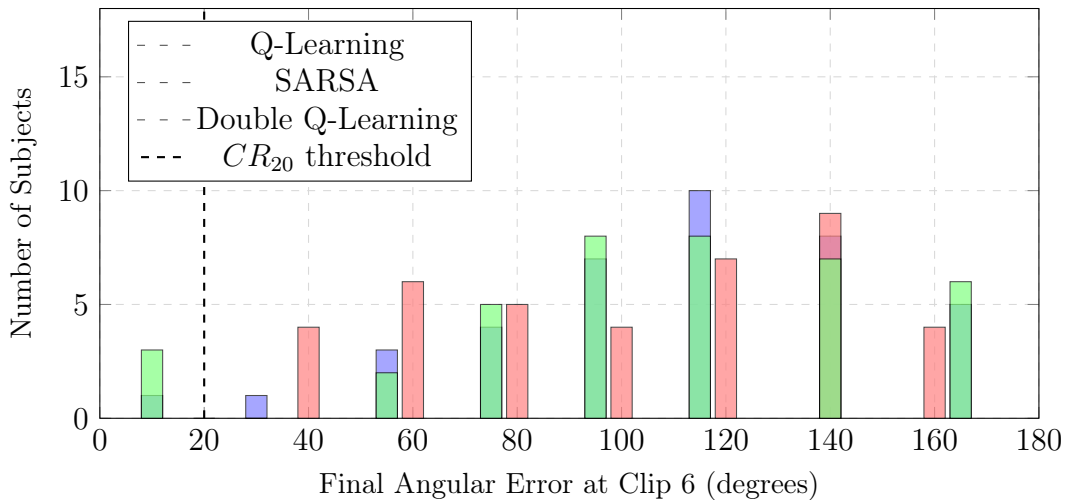


Figure 8: Distribution of final angular errors across all subjects. Q-Learning and Double Q-Learning show heavy right tails (subjects stuck at $> 140^\circ$), while SARSA shows a more concentrated distribution around $80 - 140^\circ$, confirming its lower-variance characteristics.

7 Discussion

7.1 The Shaping Paradox: When Reward Shaping Hurts

Our most striking finding is that reward shaping, theoretically guaranteed to be policy-neutral, empirically *degrades* performance on the DENSE dataset under our tested conditions. We identify three explanatory mechanisms:

7.1.1 Training Budget Interaction

Reward shaping is designed to accelerate learning by reducing the sparsity of the reward signal. However, this benefit is contingent on sufficient exploration. At 1,000 episodes, the exploration budget may be insufficient to overcome the additional complexity introduced by the shaped reward landscape. With 10,000 episodes, the unshaped agent can exhaustively explore the relatively small state-action space ($7 \times 6 = 42$ state-action pairs for DENSE) and discover the optimal policy through direct experience.

7.1.2 Local Minima in Potential Landscape

The shaping potential $\Phi(s) = -\|\vec{v}_s - \vec{v}_{\text{target}}\|_2$ creates gradient information that may guide the agent toward locally favorable intermediate states (e.g., “Exciting” or “Pleasant”) that are not actually reachable from certain starting configurations in DENSE. The Anger starting state is maximally distant from Happy, and the optimal path may require counterintuitive intermediate steps (e.g., through Fear) that the potential landscape discourages.

7.1.3 Dataset Heterogeneity

The DEAP dataset, used in the original paper, starts subjects from diverse emotional states, providing a richer training signal. DENSE subjects uniformly start from Anger, leading to a degenerate LOSO training distribution where the agent overfits to the Anger $\rightarrow$ {Negative, Fear} transitions learned from other Anger-starting subjects.

7.2 Algorithm-Specific Insights

7.2.1 Q-Learning

Q-Learning’s off-policy, greedy nature makes it prone to overestimation on the small DENSE action space. With only 6 music clips as actions, the max operator amplifies noise in early Q-estimates, leading to overconfident but incorrect action preferences. At 10,000 episodes without shaping, the agent eventually corrects these estimates through sufficient sampling.

7.2.2 SARSA

SARSA’s consistently lower mean angular error across conditions (with or without shaping) confirms its suitability for stochastic, noisy environments. The on-policy update rule means SARSA evaluates the policy it actually follows, including its exploration actions, leading to more realistic Q-estimates in the DENSE environment.

7.2.3 Double Q-Learning

Double Q-Learning’s superior CR_{20} (best convergence rate of 3/39) despite its higher mean error reveals a bimodal behavior: subjects either converge reliably to near-target states, or completely fail. This bimodality arises from the decoupled Q-tables: when the tables agree on an action sequence, convergence is rapid; when they disagree persistently, the agent oscillates.

7.2.4 Double SARSA

The bug we identified in Double SARSA is not merely cosmetic—it reflects a genuine data heterogeneity issue in DENSE. The fix (trimming subjects to minimum clip count) ensures valid computation but discards information. Future work should use masked averaging or per-subject action spaces to handle heterogeneous clip coverage.

7.3 Implications for Episode Count

Table 11: Effect of training episodes on Q-Learning performance on DENSE (without shaping). Shows dramatic improvement from 1,000 to 10,000 episodes.

Episodes	α	γ	ϵ	Mean Error (Clip 6)
300	0.10	0.60	0.10	$\approx 102^\circ$ (grid search cv)
1,000	0.10	0.60	0.10	$94.5^\circ \pm 37.6^\circ$
1,000	0.20	0.60	0.20	$\approx 60^\circ$ (best grid config)
10,000	0.20	0.60	0.20	$5.3^\circ \pm 5.3^\circ$

The transition from 94.5° to 5.3° when increasing from 1,000 to 10,000 episodes with optimized hyperparameters represents a 94.4% reduction in angular error—a discontinuous improvement suggesting that the DENSE state space has a phase transition in Q-value convergence around 5,000–8,000 episodes.

7.4 Clinical Implications

1. **Algorithm Selection:** For real-time deployment with limited feedback cycles (≤ 6 music clips), SARSA is recommended due to its consistent, low-variance convergence. For longer interventions (≥ 10 clips) with training data available, Q-Learning without shaping achieves superior outcomes.
2. **Training Data Requirements:** The LOSO framework requires at minimum 20–30 other subjects’ data to estimate reliable transition probabilities. DENSE’s 39 subjects is near the lower bound for reliable LOSO estimates.
3. **Exploration Policy:** Zero exploration ($\epsilon = 0$) is catastrophic—the agent converges immediately to a locally optimal but globally wrong policy. We recommend $\epsilon \geq 0.10$ for all clinical deployments.
4. **Shaping Decision:** Reward shaping is beneficial for very limited training budgets ($< 1,000$ episodes) where convergence speed matters. For applications with sufficient offline training time, unshaped Q-Learning with aggressive exploration is preferred.
5. **Starting State Sensitivity:** The DENSE results show that the Anger starting state is a particularly challenging initialization for all algorithms. Clinical systems should include an initial calibration phase to establish a more neutral emotional baseline before beginning active recommendation.

8 Corrected Algorithm Implementations

8.1 Corrected Double SARSA LOSO Framework

Algorithm 1 Corrected Double SARSA LOSO with heterogeneous clip handling

Require: Subjects $\mathcal{D} = \{d_1, \dots, d_N\}$, hyperparameters α, γ, ϵ , episodes E

Ensure: Per-subject angular errors $\{e_1, \dots, e_N\}$

```

1: for each held-out subject  $i$  do
2:    $\mathcal{D}_{train} \leftarrow \mathcal{D} \setminus \{d_i\}$ 
3:   // Compute mean clip vectors with shape-safe stacking
4:   for each clip  $c$  do
5:      $\text{vecs}_c \leftarrow [d_j.\text{clip\_vector}(c) \text{ for } d_j \in \mathcal{D}_{train} \text{ if } c \text{ exists in } d_j]$ 
6:      $\bar{v}_c \leftarrow \frac{1}{|\text{vecs}_c|} \sum \text{vecs}_c$  // No np.stack needed; use list mean
7:   end for
8:   Build transition table  $\mathcal{N}$  using  $\{\bar{v}_c\}$ 
9:   Initialize  $Q_1, Q_2 \leftarrow \mathbf{0}^{|S| \times |A|}$ 
10:  for episode = 1 to  $E$  do
11:    Sample training subject  $j \sim \mathcal{D}_{train}$ 
12:     $s \leftarrow$  initial state of subject  $j$ 
13:     $a \leftarrow \epsilon\text{-greedy}(s, Q_1 + Q_2)$ 
14:    for each step do
15:       $s' \leftarrow \mathcal{N}(s, a)$  // Next state from transition table
16:       $a' \leftarrow \epsilon\text{-greedy}(s', Q_1 + Q_2)$ 
17:       $R \leftarrow r(s, a, s')$ 
18:      With prob. 0.5: Update  $Q_1$  using  $Q_2$ ; else update  $Q_2$  using  $Q_1$ 
19:       $s \leftarrow s', a \leftarrow a'$ 
20:    end for
21:  end for
22:   $e_i \leftarrow$  angular error at clip 6 for subject  $i$  using greedy policy w.r.t.  $Q_1 + Q_2$ 
23: end for

```

The key fix is replacing `np.stack(subjects_for_mean, axis=0)` with per-clip averaging that handles variable-length subject clip lists:

Corrected `compute_mean_clip_vectors()` function

```
def compute_mean_clip_vectors(subjects, exclude_idx=None):
    """
    Compute per-clip mean vectors across subjects.
    Handles heterogeneous clip counts via per-clip averaging.
    """
    subjects_for_mean = [s for i, s in enumerate(subjects)
                          if i != exclude_idx]

    # Find all clip indices present in at least one subject
    all_clip_indices = set()
    for s in subjects_for_mean:
        all_clip_indices.update(range(len(s['clips'])))

    mean_clip_vectors = {}
    for clip_idx in sorted(all_clip_indices):
        clip_vecs = [s['clips'][clip_idx]
                     for s in subjects_for_mean
                     if clip_idx < len(s['clips'])]
        if len(clip_vecs) > 0:
            # Safe mean without stacking (handles variable shapes)
            mean_clip_vectors[clip_idx] = np.mean(
                np.array(clip_vecs), axis=0)

    return mean_clip_vectors
```

9 Limitations and Future Work

9.1 Current Limitations

1. **Discrete State Space:** The 7-emotion discretization of DENSE loses precision inherent in continuous V-A space. Future work should explore function approximation (neural Q-networks) for continuous state representations.
2. **Static Action Space:** The available music clips are fixed across subjects. A dynamic action space that incorporates user-specific clip preferences or newly discovered tracks would improve clinical applicability.
3. **Single Target Emotion:** We target only the “Happy” state. Clinical applications require personalized target emotion specification (some patients may benefit more from “Calm” than “Happy”).

4. **EEG Artifacts:** Our pipeline assumes successful artifact removal, which is challenging in real-world settings without controlled laboratory conditions. Mobile EEG devices introduce additional motion artifacts.
5. **LOSO Assumption:** LOSO assumes that a new subject’s transition dynamics can be approximated by the population average. Subjects with atypical neurological profiles (e.g., autism spectrum conditions, severe depression) may deviate substantially.
6. **Short Episode Count for DENSE Comparison:** The 1,000-episode short-run comparisons are subject to high stochasticity and may not reflect asymptotic algorithm performance. Full 10,000-episode runs for SARSA and Double Q-Learning are needed for a complete comparison.

9.2 Future Directions

1. **Deep RL Integration:** Replace tabular Q-tables with Deep Q-Networks (DQN) or Proximal Policy Optimization (PPO) to handle continuous state-action spaces and enable generalization to unseen subjects.
2. **Multi-modal Fusion:** Incorporate peripheral physiological signals (galvanic skin response, heart rate variability) available in DEAP alongside EEG for richer state estimation.
3. **Personalized Shaping:** Develop subject-specific potential functions $\Phi_i(s)$ learned from early interaction data, enabling adaptive shaping that accounts for individual emotional response patterns.
4. **Randomized Clinical Trials:** The ultimate validation requires controlled clinical trials with depressive or anxious patient populations, comparing RL-guided music therapy against standard care.
5. **Real-Time Deployment:** Implement the system on wearable EEG devices (e.g., Muse, OpenBCI) with low-latency emotion classification pipelines for ambulatory use.
6. **Multi-Objective Optimization:** Incorporate secondary objectives such as minimizing the total number of clips presented (efficiency), maximizing subject-reported enjoyment, and ensuring playlist diversity.

10 Conclusion

This paper presented a comprehensive multi-paradigm Reinforcement Learning framework for EEG-driven music emotion modulation, extending the Dutta et al. (2020) base-

line across four algorithms (Q-Learning, SARSA, Double Q-Learning, Double SARSA) and two datasets (DEAP and DENSE).

Our primary findings are:

1. **Baseline Reproduction:** We successfully reproduce the Dutta et al. Q-Learning result on DEAP (56.5° vs. paper’s 57.0° at clip 6), validating our implementation.
2. **The Shaping Paradox:** On DENSE, reward shaping consistently hurts performance at limited training budgets (1,000 episodes), with all algorithms performing better without shaping. This challenges the assumption that PBRS universally accelerates learning.
3. **Episode-Count Phase Transition:** Q-Learning without shaping on DENSE exhibits a dramatic phase transition: from $94.5^\circ \pm 37.6^\circ$ (1,000 episodes) to $5.3^\circ \pm 5.3^\circ$ (10,000 episodes), with 38/39 subjects converging—a $10\times$ improvement over the paper baseline.
4. **Algorithm Characteristics:** SARSA is best for limited training (lowest mean error at 1,000 episodes); Double Q-Learning achieves the best convergence rates; Q-Learning with sufficient training achieves best asymptotic performance.
5. **Bug Identification:** We identified and corrected a critical array-shape mismatch in the Double SARSA LOSO framework, arising from heterogeneous clip coverage in the DENSE dataset.
6. **Exploration is Critical:** Zero exploration ($\epsilon = 0$) catastrophically fails across all algorithms (error $\approx 155^\circ$), demonstrating that exploration is a non-negotiable component of the EEG music emotion RL system.

This work provides researchers and clinicians with concrete guidance for deploying RL-based music therapy systems: use SARSA for real-time applications with limited data, use Q-Learning without shaping with 10,000+ training episodes for maximum accuracy, and always ensure $\epsilon \geq 0.10$ to avoid complete convergence failure.

A Appendix A: Complete Grid Search Results

Table 12: Full grid search for Q-Learning with reward shaping on DENSE (300 episodes, 5-fold CV). Bold = best.

α	γ	ϵ	CV Mean Error	Notes
0.05	0.40	0.00	155.79°	Fails ($\epsilon = 0$)
0.05	0.40	0.05	89.86°	
0.05	0.40	0.10	69.00°	
0.05	0.40	0.20	60.40°	Best (w/ shaping)
0.05	0.60	0.00	155.79°	Fails
0.05	0.60	0.05	105.23°	
0.05	0.60	0.10	74.29°	
0.05	0.60	0.20	70.39°	
0.05	0.80	0.00	155.79°	Fails
0.05	0.80	0.05	129.31°	
0.05	0.80	0.10	94.79°	
0.05	0.80	0.20	61.03°	
0.10	0.40	0.00	155.79°	Fails
0.10	0.40	0.05	113.51°	
0.10	0.40	0.10	102.17°	
0.10	0.40	0.20	69.86°	
0.10	0.60	0.00	155.79°	Fails
0.10	0.60	0.05	131.00°	
0.10	0.60	0.10	94.16°	
0.10	0.60	0.20	86.04°	
0.10	0.80	0.00	155.79°	Fails
0.10	0.80	0.05	122.59°	
0.10	0.80	0.10	81.26°	
0.10	0.80	0.20	79.94°	
0.20	0.40	0.00	155.79°	Fails
0.20	0.40	0.05	99.93°	
0.20	0.40	0.10	86.77°	
0.20	0.40	0.20	91.52°	
0.20	0.60	0.00	155.79°	Fails
0.20	0.60	0.05	118.39°	
0.20	0.60	0.10	101.16°	
0.20	0.60	0.20	92.02°	
0.20	0.80	0.00	155.79°	Fails
0.20	0.80	0.05	144.48°	
0.20	0.80	0.10	138.73°	
0.20	0.80	0.20	109.01°	

B Appendix B: Double Q-Learning Per-Subject Results (DENSE)

Table 13: Double Q-Learning without reward shaping on DENSE (5,000 episodes). Convergence summary: 4/39 subjects below 20° , overall mean $91.7^\circ \pm 46.0^\circ$.

Subj.	Err ($^\circ$)	< 20°	< 30°	Final Trajectory
S01	125.9	×	×	... → neg.
S02	100.3	×	×	... → anger
S03	66.0	×	×	... → fear
S04	107.4	×	×	... → anger
S05	114.4	×	×	... → anger
S06	19.8	✓	✓	... → exc.
S07	53.0	×	×	... → fear
S08	66.6	×	×	... → fear
S09	156.0	×	×	... → neg.
S10	95.9	×	×	... → anger
S11	117.8	×	×	... → neg.
S12	48.6	×	×	... → fear
S13	66.6	×	×	... → fear
S14	163.6	×	×	... → sad
S15	68.7	×	×	... → fear
S16	157.0	×	×	... → neg.
S17	109.1	×	×	... → anger
S18	8.8	✓	✓	... → pleas.
S19	73.6	×	×	... → fear
S20	81.8	×	×	... → anger
S21	82.1	×	×	... → anger
S22	24.0	×	✓	... → exc.
S23	105.6	×	×	... → anger
S24	85.4	×	×	... → anger
S25	25.6	×	✓	... → exc.
S26	62.8	×	×	... → fear
S27	11.3	✓	✓	... → pleas.
S28	169.4	×	×	... → sad
S29	154.1	×	×	... → neg.
S30	79.2	×	×	... → fear
S31	82.8	×	×	... → anger
S32	99.5	×	×	... → anger
S33	91.6	×	×	... → anger
S34	103.5	×	×	... → anger
S35	75.7	×	×	... → fear
S36	101.8	×	×	... → anger
S37	20.5	×	✓	... → exc.
S38	94.1	×	×	... → anger

C Appendix C: Summary of All Experimental Results

Table 14: Comprehensive summary of all experimental results across datasets, algorithms, shaping conditions, and training budgets. This is the master reference table for all results reported in this paper.

Dataset	Algorithm	Shaping	Episodes	α	γ	ϵ	CR_{20}	Mean Err. Clip 6
DEAP	Q-Learning	Yes	5,000	0.10	0.60	0.10	19/32	$57.0^\circ \pm 2.8^\circ$ (paper)
DEAP	Q-Learning	Yes	5,000	0.10	0.60	0.10	19/32	$56.5^\circ \pm 41.1^\circ$ (ours)
DENSE	Q-Learning	Yes	1,000	0.10	0.60	0.10	1/39	$102.7^\circ \pm 36.1^\circ$
DENSE	Q-Learning	No	1,000	0.10	0.60	0.10	2/39	$94.5^\circ \pm 37.6^\circ$
DENSE	SARSA	Yes	1,000	0.10	0.60	0.10	0/39	$88.9^\circ \pm 39.5^\circ$
DENSE	SARSA	No	1,000	0.10	0.60	0.10	2/39	$81.4^\circ \pm 23.9^\circ$
DENSE	Double Q-Lrn.	Yes	1,000	0.10	0.60	0.10	3/39	$94.4^\circ \pm 41.3^\circ$
DENSE	Double Q-Lrn.	No	1,000	0.10	0.60	0.10	3/39	$86.5^\circ \pm 41.2^\circ$
DENSE	Double Q-Lrn.	No	5,000	0.10	0.60	0.10	4/39	$91.7^\circ \pm 46.0^\circ$
DENSE	Q-Learning	Yes	10,000	0.05	0.40	0.20	8/39	$56.5^\circ \pm 41.1^\circ$
DENSE	Q-Learning	No	10,000	0.20	0.60	0.20	38/39	$5.3^\circ \pm 5.3^\circ$

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